Chapter 2 • Logarithms

Solved Exercise 2.1

1. Express the numbers in scientific notation.
(i) 2000000 = 2×1000000
= 2×10^6.
(ii) 48900 = 4.89×10000
= 4.89×10^4.
(iii) 0.0042 = 4.2×0.001
= 4.2×10^{-3}.
(iv) 0.000009 = 9×0.000001
= 9×10^{-6}.
(v) 73×10^3 = 7.3×10×10^3
= 7.3×10^4.
(vi) 0.65×10^2 = 65
= 6.5×10^1.

2. Express the numbers in ordinary notation.
(i) 8.04×10^2 = 8.04×100
= 804.
(ii) 3×10^5 = 3×100000
= 300,000.
(iii) 1.5×10^{-2} = 1.5×0.01
= 0.015.
(iv) 1.77×10^7 = 1.77×10000000
= 17,700,000.
(v) 5.5×10^{-6} = 5.5×0.000001
= 0.0000055.
(vi) 4×10^{-5} = 4×0.00001
= 0.00004.

3. The speed of light is approximately 3 × 10^8 $\frac{m}{s}$ Express it in standard form.
A positive exponent of 8 moves the decimal point eight places right: 3 × 10^8 = 300,000,000.
300,000,000 $\frac{m}{s}$

4. Express Earth's equatorial circumference, 40,075,000 m, in scientific notation.
Move the decimal seven places left to obtain 4.0075.
4.0075 × 10^7 m.

5. The diameter of Mars is 6.779 × 10^3 km. Express it in standard form.
6.779 × 1000 = 6779.
6,779 km.

6. The diameter of Earth is about 1.2756 × 10^4 km. Express it in standard form.
1.2756 × 10,000 = 12,756.
12,756 km.

studysapiens.com